

ResQAgent — Project Presentation & Jury Defense

AUTONOMOUS AGENTIC AI EMERGENCY RESPONSE & COORDINATION SYSTEM

Target Score: 100 / 100 Marks	Architecture: 6-Agent Autonomous Cognitive Loop	Status: Live & Operational on Port 8000
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Executive Pitch for Jury: ResQAgent is an autonomous emergency response coordination platform powered by an end-to-end Agentic AI cognitive cycle (PERCEIVE → REASON → DECIDE → ACT → OBSERVE → ADAPT → REPORT). It eliminates 911 dispatch latency by clinically triaging distress messages, optimizing field responder dispatch via multi-factor proximity/specialization scoring, monitoring 30-second SLA acknowledgment timeouts, and autonomously re-routing missions with zero human bottleneck. Backed by 17 automated tests, production RBAC, and Gemini 1.5 Flash.

1. AGENTIC AI EVALUATION SCORECARD (100 MARKS)

Metric	Marks	Focus Area	ResQAgent Compliance & Implementation
1. Problem Understanding & Identification	10	Clearly identify real problem, users & needs	Golden Hour dispatch latency; 4 distinct personas (Citizen, Responder, Dispatch, Admin)
2. Innovation, Creativity & Use-Case	10	Original & meaningful solution	Self-healing dispatch loop; autonomous SLA timeout recovery; explainable cognitive actions
3. Agent Architecture, Workflow & Reasoning	20	Task breakdown, reasoning & workflows	6 specialized agents; state machine: PERCEIVE → REASON → DECIDE → ACT → OBSERVE → ADAPT
4. AI Implementation & Technical Stack	20	LLMs, prompting, tool calling, APIs	Gemini 1.5 Flash REST API; structured JSON schema; deterministic cognitive fallback engine
5. Functionality, Performance & Reliability	15	Reliable real-world execution	17-test security suite; bcrypt 12-round hashing; JWT tokens; database-level privacy isolation
6. GitHub, Code Quality & Deployment	5	Clean code, README & deployment	Decoupled architecture; comprehensive README; 1-click startup on http://127.0.0.1:8000
7. Profile Building (LinkedIn & Instagram)	5	Professional public showcase	Viral technical showcase post, architectural diagrams, and carousel copy provided
8. Presentation & Live Demonstration	15	Explain & demonstrate working app	4-minute live demo script showing SOS intake, automated dispatch, SLA breach & report
TOTAL	100	Comprehensive Project Coverage	Fully Operational & Validated Across All Criteria

METRIC 1: PROBLEM UNDERSTANDING & IDENTIFICATION (10 MARKS)

The Real-World Emergency Crisis: In trauma medicine, survival drops exponentially after the 'Golden Hour' (the first 60 minutes post-incident). Traditional 911 CAD systems suffer from three acute bottlenecks:

- **Dispatcher Overload & Fatigue:** In mass-casualty events, operators face severe cognitive overload, leading to delayed categorization.

- **Unmonitored SLA Dropouts:** If a responder is delayed, has radio failure, or declines, traditional systems stay idle until someone notices 5–10 minutes later.
- **Fragmented Communications:** Callers are left in the dark without arrival updates, and trauma hospitals receive zero pre-arrival casualty heads-up.

4 Distinct Real-World Stakeholder Personas Addressed:

1. **Citizen / Caller:** One-touch distress SOS submission, live arrival tracking, privacy protection, and reassuring automated SMS updates.
2. **Field Responder (EMT/Fire/Police):** Mobile alerts with exact clinical priority (P1-P4), patient conditions, and 1-click status transitions.
3. **Central Dispatcher:** Full operational oversight across all live incidents, responder fleet map, and transparent agent reasoning traces.
4. **System Administrator:** Role-based access provisioning, security audits, simulation triggers, and user lifecycle governance.

METRIC 2: INNOVATION, CREATIVITY & USE-CASE RELEVANCE (10 MARKS)

Why ResQAgent is an Original, Breakthrough Solution:

- **Not a Chatbot — An Autonomous Actuator:** Most hackathon submissions wrap LLMs into text chat interfaces. ResQAgent is an autonomous decision and actuation engine that changes database states, reserves units, tracks timers, and sends multi-party alerts.
- **Self-Healing Dynamic Escalation:** If an assigned unit fails to respond within 30 seconds, the agent autonomously blacklists that responder, re-engages the selection algorithm, and re-routes the next best unit with zero human intervention.
- **Full Cognitive Transparency:** Every decision made by any agent is persisted in an auditable *AgentAction* log detailing agent name, action type, raw input, clinical/mathematical reasoning, and output.

METRIC 3: AGENT ARCHITECTURE, WORKFLOW & REASONING DESIGN (20 MARKS)

ResQAgent separates concerns into 6 specialized autonomous agents coordinating within a strict state machine:

Agent Name	Cognitive Phase	Core Responsibility & Operational Logic
1. EmergencyAnalysisAgent	PERCEIVE & REASON	Parses unstructured text; classifies type (Accident, Fire, Medical, Crime, Rescue), severity (Critical, High, Med, Low), and priority (P1–P4).
2. ResponderSelectionAgent	DECIDE	Evaluates available fleet using mathematical utility score: $Score = (10 / Dist) * Spec_Bonus(1.5x)$. Matches units to emergency type.
3. CommunicationAgent	ACT	Omnichannel multi-stakeholder alert dispatch: App push to Responder, SMS reassurance to Citizen, Radio log to Dispatch, Secure Alert to Hospital.
4. MonitoringAgent	OBSERVE	Enforces strict 30-second acknowledgment SLA window. Tracks mission progression and flags timeouts.
5. EscalationAgent	ADAPT	Self-healing loop: Blacklists dropped unit, searches candidate pool excluding failed units, assigns new responder. Safeguard: Max 3 attempts.
6. AllIncidentReportAgent	REPORT	Synthesizes post-incident debrief dossier from timeline logs, response times, and escalations. Scores agentic performance.

METRIC 4: AI IMPLEMENTATION & TECHNICAL STACK (20 MARKS)

- **LLM Integration:** Powered by Google Gemini 1.5 Flash via REST API (backend/agents/llm_service.py) using strict system prompts and JSON mime type enforcement.
- **Mission-Critical Dual-Engine Architecture:** If internet connectivity is interrupted or LLM API rate limits are hit, the system triggers an automatic **Deterministic Cognitive Fallback Engine** (backend/agents/analysis_agent.py lines 47-98). Distress calls are never lost or hung.
- **RESTful Tool Calling & Micro-APIs:** 8 modular FastAPI controllers handling Authentication, Administration, Incidents, Responders, Assignments, Simulation, Users, and Reports.

METRIC 5: FUNCTIONALITY, PERFORMANCE & RELIABILITY (15 MARKS)

- **Cryptographic RBAC Security:** Passwords hashed with bcrypt (12 salt rounds). Plaintext passwords never stored. Authenticated via signed HS256 JWT tokens.
- **Data Isolation:** Citizens cannot view incidents from other citizens. Responders can only access missions assigned to them. Administrative endpoints reject unauthorized roles with HTTP 403.

• 3 Automated Verification Suites:

1. backend/test_auth_suite.py: 17 comprehensive security and RBAC test scenarios.
2. backend/test_api_suite.py: End-to-end REST endpoint testing across all routers.
3. backend/test_agent_flow.py: Full autonomous cognitive loop & SLA escalation verification.

METRIC 6: GITHUB, CODE QUALITY & DEPLOYMENT (5 MARKS)

- Clean modular repository separating backend agents, models, API routes, and pure vanilla reactive frontend.
- Fully documented setup in README.md with clear role descriptions, API specifications, and test commands.
- Live server runs with single command: .\venv\Scripts\python -m uvicorn backend.main:app --host 127.0.0.1 --port 8000

METRIC 7: LINKEDIN & INSTAGRAM PROFILE BUILDING (5 MARKS)

LinkedIn Post Template (Ready to Publish):

■ Can Agentic AI save lives during the 'Golden Hour' of emergency response?
Thrilled to introduce ResQAgent — an autonomous Agentic AI Emergency Coordination platform. Unlike basic chatbots, ResQAgent executes a 6-stage cognitive loop (Perceive → Reason → Decide → Act → Observe → Adapt → Report). It eliminates 911 dispatch latency, scores responders by proximity & medical specialization, and self-heals by re-routing assignments when SLA timeouts occur. Built with Python, FastAPI, Gemini 1.5 Flash & 17 automated security tests! #AgenticAI #Python #FastAPI #AI

METRIC 8: PRESENTATION & LIVE DEMONSTRATION PLAYBOOK (15 MARKS)

Step-by-Step 4-Minute Live Demo Script:

1. **Minute 0:00 - 1:00 (Citizen SOS Intake):** Open http://localhost:8000, enter accident distress call ('Severe crash, car smoking, driver trapped & bleeding'). Show status transitioning from ANALYZING to ANALYZED with P1-Critical clinical priority.
2. **Minute 1:00 - 2:00 (Automated Selection & Multi-Alerts):** Show SelectionAgent assigning unit Suresh (0.8km away). Show CommunicationAgent sending push to responder, SMS to citizen, and trauma advisory to hospital.
3. **Minute 2:00 - 3:00 (The WOW Factor: SLA Escalation):** Click 'Simulate SLA Timeout'. Watch MonitoringAgent detect timeout, EscalationAgent blacklist Suresh, and reassign unit Ravi (1.5km) with citizen reassurance.

4. **Minute 3:00 - 4:00 (Resolution & AI Debrief):** Mark mission Completed. Click 'Generate AI Incident Report'. Show executive debrief dossier and 95/100 performance score.

JURY Q&A; DEFENSE CHEAT SHEET

Jury Question	Winning High-Impact Answer
Why multi-agent instead of one prompt?	Emergency dispatch requires deterministic mathematical scoring (distance, availability) and strict SLA timing, which monolithic LLM prompts cannot reliably guarantee. Our multi-agent separation ensures speed, precision, and zero hallucinations.
What if Gemini API goes down?	ResQAgent features a built-in Deterministic Cognitive Fallback Engine. If the LLM API fails or times out, rule-based clinical keyword extraction takes over in milliseconds without dropping the call.
How do you prevent infinite loops?	EscalationAgent has a hardcoded safeguard: MAX_ESCALATION_ATTEMPTS = 3. If 3 automated reassignment attempts fail, the incident immediately escalates to human central dispatch (ESCALATED_TO_DISPATCH).
How is citizen privacy protected?	Full RBAC with bcrypt 12-round password hashing and JWT tokens. Database queries are scoped so citizens can only view their own records. Non-admin access to admin routes returns HTTP 403.